Graphics in R using ggplot

ECO1400

Install and load required R packages

```
chooseCRANmirror(ind=0)
if (!require("xts")) install.packages("xts") # Time series analysis functions
if (!require("ggplot2")) install.packages("ggplot2") # Plot functions
if (!require("stats")) install.packages("stats") # Statistical functions
library(xts)      # functions: xts
library(ggplot2)  # functions: ggplot
library(stats)    # functions: decompose
```

Load data (5-Year Treasury Constant Maturity Rate (Source))

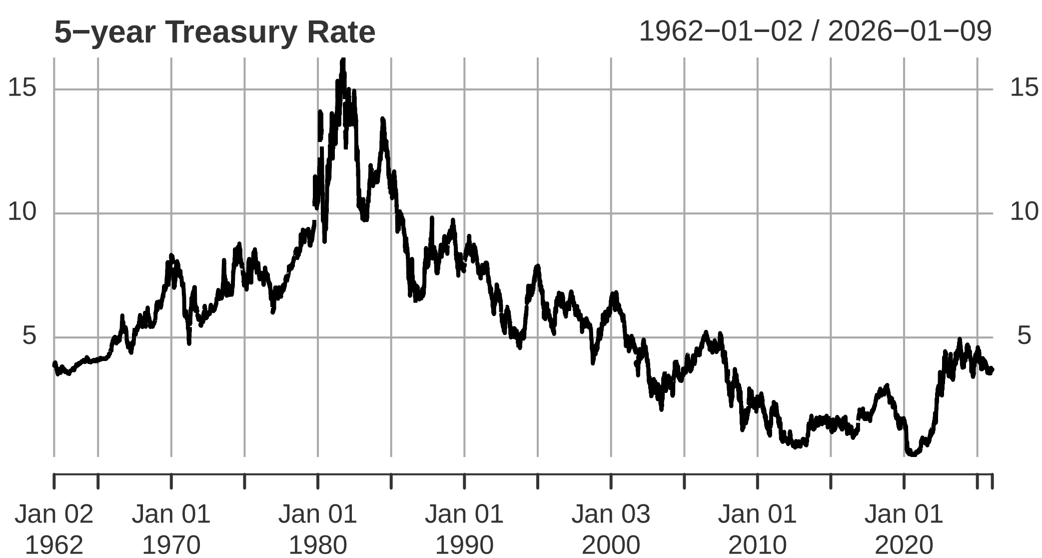
```
if (!require("quantmod")) install.packages("quantmod")
library(quantmod) # functions: getSymbols
DGS5.table <- getSymbols("DGS5", src="FRED", return.class="xts", auto.assign=FALSE)
```

1. Base Graphics

In R there are many ways to do one thing, especially when it comes to graphics. R comes with built-in graphics functionality, typically referred to as "base graphics". For line plots, the main function is `plot()`. Its output style depends on the class of the object passed in as argument.

Plot of an *xts* class object

```
plot(DGS5.table, main="5-year Treasury Rate")
```

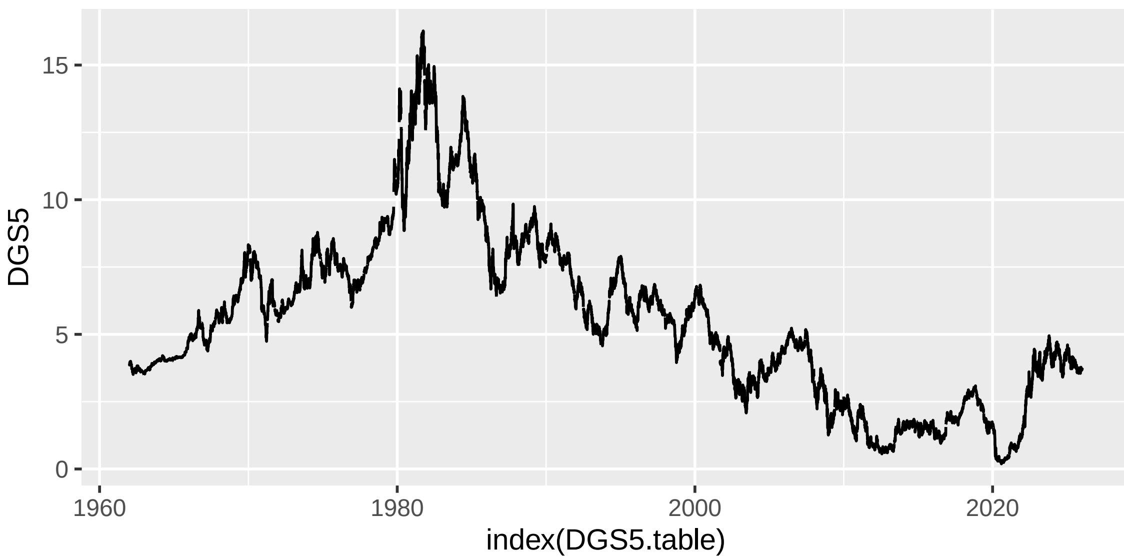


2. ggplot

The `ggplot2` package contains graphics functions that extend the based graphics environment. First, we specify the data to be plotted, and the plot "aesthetics" in the `aes()` argument such as specifying the X and Y axis variables. Next, the plot is activated by adding one or more "layers" (also called "geoms") separated by the plus sign `+`.

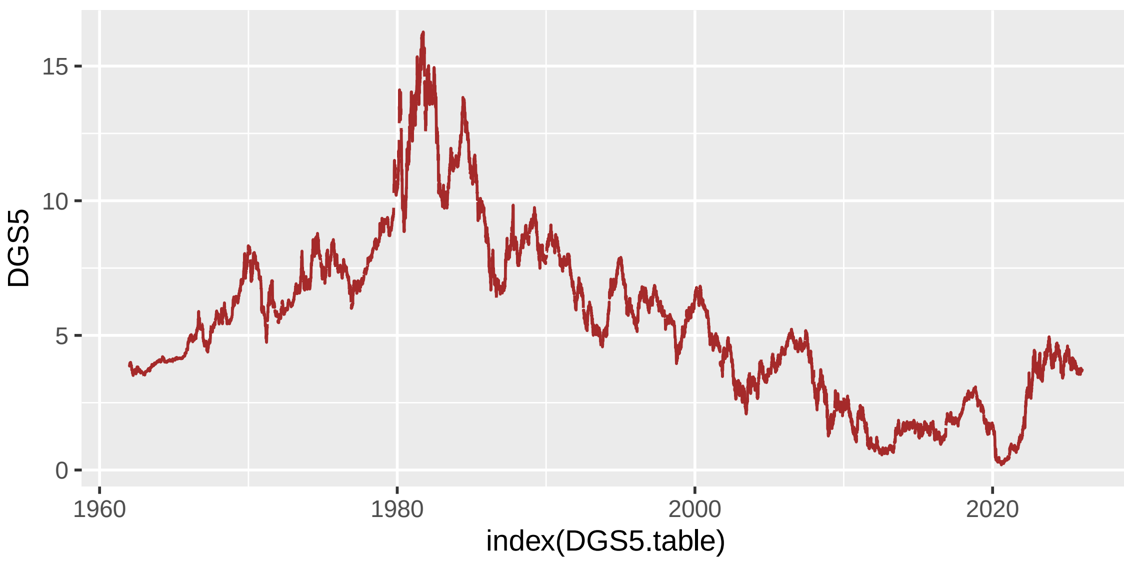
The `geom_line()` layer activates a line plot.

```
ggplot(data=DGS5.table, aes(x=index(DGS5.table), y=DGS5)) + geom_line()
```



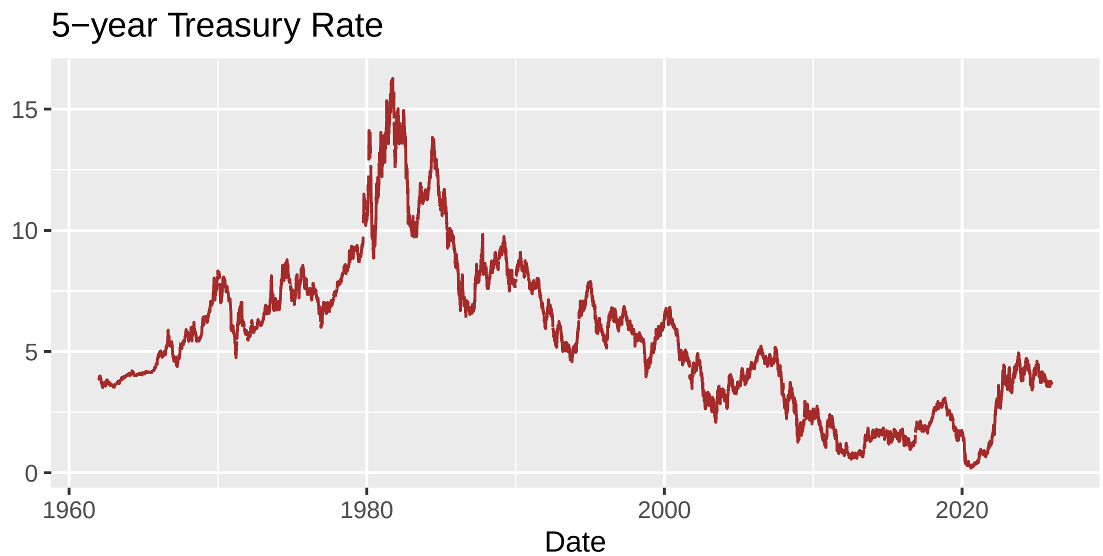
The `geom_line()` layer also contains options for line plot colour and type.

```
ggplot(data=DGS5.table, aes(x=index(DGS5.table), y=DGS5)) +
  geom_line(color="brown",linetype="solid")
```



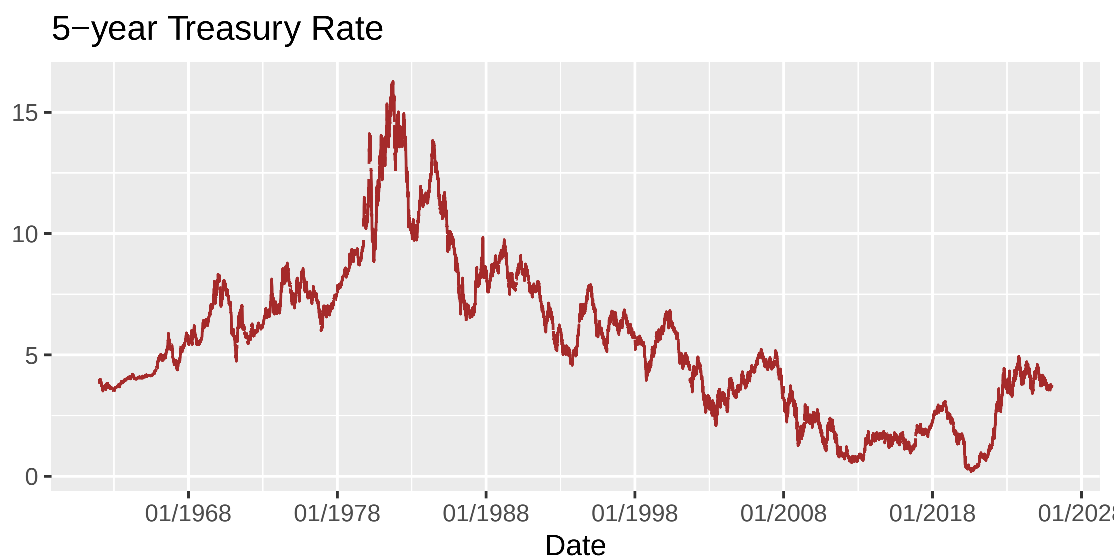
The `labs()` layer specifies the x,y labels and the graph title.

```
ggplot(data=DGS5.table, aes(x=index(DGS5.table), y=DGS5)) +
  geom_line(color="brown",linetype="solid") +
  labs(x="Date", y="", title="5-year Treasury Rate")
```



The `scale_x_date()` layer specifies the frequency and format of the date labels on the x-axis.

```
ggplot(data=DGS5.table, aes(x=index(DGS5.table), y=DGS5)) +
  geom_line(color="brown",linetype="solid") +
  labs(x="Date", y="", title="5-year Treasury Rate") +
  scale_x_date(date_breaks="10 years", date_labels = "%m/%Y")
```



The `theme_minimal()` layer changes the overall theme (i.e. design) of the graph.

```
ggplot(data=DGS5.table, aes(x=index(DGS5.table), y=DGS5)) +
  geom_line(color="brown",linetype="solid") +
  labs(x="Date", y="", title="5-year Treasury Rate") +
  scale_x_date(date_breaks="10 years", date_labels = "%m/%Y") +
  theme_minimal()
```

The `theme()` layer further modifies the components of a theme. For example, we can change the size of the graph title to 10 pts.

```
ggplot(data=DGS5.table, aes(x=index(DGS5.table), y=DGS5)) +
  geom_line(color="brown",linetype="solid") +
  labs(x="Date", y="", title="5-year Treasury Rate") +
  scale_x_date(date_breaks="10 years", date_labels = "%m/%Y") +
  theme_minimal() +
  theme(plot.title = element_text(size=10))
```

3. Graph Size in RStudio

The size of graphs in RStudio output can be changed by the options `fig.width=` and `fig.height=` in the code chung headers. For example, we can generate a 6x3 inch graph as follows.

```
```{r, fig.width=6, fig.height=3}
<plot functions>
```
```

4. ggplot Reference

A useful reference graph gallery with various types of ggplot layers can be found here.